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Battery pack having structure allowing input of cooling water into battery module upon occurrence of thermal runaway phenomenon and ESS comprising same

Abstract

A battery pack includes a pack housing; a plurality of battery modules; a water tank connected to the plurality of battery modules; a coolant tube including a main tube connected to the water tank, a plurality of supply tubes configured to connect the main tube and the battery modules to each other, and a bypass tube connected to the main tube at a point above a supply tube located at a top end among the plurality of supply tubes and at a point below a supply tube located at a bottom end; at least one sensor installed in the pack housing to detect a thermal runaway phenomenon generated in at least a part of the plurality of battery modules; and a controller configured to output a control signal for introducing a coolant into the battery module through the coolant tube when a thermal runaway phenomenon is detected by the sensor.

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Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates to a battery pack having a structure for introducing a coolant into a battery module when a thermal runaway phenomenon occurs, and an energy storage system

(ESS) including the battery pack, and more particularly, to a battery pack having a structure capable of preventing a thermal runaway phenomenon from spreading to neighboring battery modules by efficiently introducing a coolant into battery modules in which problem occurs, when the risk of occurrence of a thermal runaway phenomenon is detected in two or more battery modules among a plurality of battery modules of the battery pack, and ESS including the battery pack.

(2) The present application claims priority to Korean Patent Application No. 10-2020-0027903 filed on Mar. 5, 2020 in the Republic of Korea, the disclosures of which are incorporated herein by reference.

BACKGROUND ART

(3) In a battery module that includes a plurality of battery cells, if an abnormality such as a short circuit occurs in some battery cells to raise temperature continuously so that the temperature of the battery cell exceeds a critical temperature, a thermal runaway phenomenon occurs. If a thermal runaway phenomenon occurs in some battery cells as described above, safety issues may be generated.

(4) If a flame or the like is generated due to the thermal runaway phenomenon occurring in some battery cells, the flame rapidly raises the temperature of adjacent battery cells, and thus the thermal runaway phenomenon may be propagated to adjacent cells within a short time.

(5) Eventually, if the thermal runaway phenomenon occurring in some battery cells is not quickly responded, it may lead to disasters such as ignition and explosion of a battery module or a battery pack, which is a battery unit with a greater capacity than the battery cell, and this may not only result in property damage but also cause safety problems.

(6) Thus, if a flame occurs due to the thermal runaway phenomenon in some battery cells inside the battery module, it is urgently necessary to quickly lower the temperature inside the battery module to prevent the flame from spreading further.

(7) Moreover, a battery module adopting an air-cooled structure has an air channel through which a coolant leaks without staying inside even though the coolant is introduced to lower the temperature inside the battery module and extinguish the flame. Thus, it is demanded to develop a battery pack having a structure capable of blocking the air channel when a coolant is introduced into a battery module where a thermal runaway phenomenon has occurred.

DISCLOSURE

Technical Problem

(8) The present disclosure is designed to solve the problems of the related art, and therefore the present disclosure is directed to preventing a flame from spreading greatly by rapidly lowering the temperature inside a battery module when the flame is generated in some battery cells in the battery module due to a thermal runaway phenomenon.

(9) However, the technical problem to be solved by the present disclosure is not limited to the above, and other objects not mentioned herein will be understood from the following description by those skilled in the art.

Technical Solution

(10) In one aspect of the present disclosure, there is provided a battery pack, comprising: a pack housing; a plurality of battery modules stacked in the pack housing; a water tank connected to the plurality of battery modules and configured to store a coolant; a coolant tube including a main tube connected to the water tank, a plurality of supply tubes configured to connect the main tube to each of the battery modules, and a bypass tube connected to the main tube at a first point above a supply tube located at a top end among the plurality of supply tubes and at a second point below a supply tube located at a bottom end among the plurality of supply tubes; at least one sensor installed in the pack housing to detect a thermal runaway phenomenon generated in at least a part of the plurality of battery modules; and a controller configured to output a control signal for introducing a coolant into the battery module through the coolant tube when a thermal runaway phenomenon is detected

by the at least one sensor.
(11) Each of the plurality of battery modules battery module may include a cell stack formed by stacking a plurality of battery cells; a module housing configured to accommodate the cell stack; an air inlet formed through the module housing at a first side of the cell stack in a stacking direction; and an air outlet formed through the module housing at a second side of the cell stack in the stacking direction.
(12) Each of the plurality of battery modules may include an expansion pad disposed inside each of the air inlet and the air outlet and configured to expand due to contact with the coolant introduced into the battery module to close the air inlet and the air outlet.
(13) Each of the plurality of battery modules may include a pair of bus bar frames respectively coupled to the first side and the second side of the cell stack.
(14) The air inlet and the air outlet may be formed at locations corresponding to an empty space formed between the bus bar frame and the module housing.
(15) The supply tube may pass through the module housing from the first side of the cell stack to the second side and communicate with an empty space formed between the bus bar frame and the module housing.
(16) The battery pack may include a plurality of valves respectively installed in the plurality of supply tubes, and the plurality of valves may be respectively installed adjacent to the plurality of battery modules to individually allow or block the flow of coolant flowing into the plurality of battery modules.
(17) The at least one sensor may be installed to each of the plurality of battery modules.
(18) The controller may output the control signal to open a valve installed adjacent to a battery module of the plurality of battery modules where a thermal runaway phenomenon is detected by the at least one sensor, among the plurality of valves.
(19) The battery pack may include a barrier installed in each of the plurality of supply tubes.
(20) The barrier may be broken broken when a temperature of the battery module exceeds a reference temperature.
(21) Meanwhile, an energy storage system (ESS) according to an embodiment of the present disclosure comprises a plurality of battery packs according to the present disclosure.
Advantageous Effects
(22) According to one aspect of the present disclosure, when a flame is generated in some battery cells inside the battery module due to a thermal runaway phenomenon, it is possible to prevent the flame from spreading further by lowering the temperature inside the battery module quickly.
(23) In addition, according to another aspect of the present disclosure, in a battery pack including an air-cooled battery module, when a coolant is introduced into the battery module where a thermal runaway phenomenon occurs, it is possible to effectively prevent the thermal runaway phenomenon from propagating by adopting a structure in which the air channel for cooling is blocked so that the coolant stays inside the battery module.

Description

DESCRIPTION OF DRAWINGS
(1) The accompanying drawings illustrate a preferred embodiment of the present disclosure and together with the foregoing disclosure, serve to provide further understanding of the technical features of the present disclosure, and thus, the present disclosure is not construed as being limited to the drawing.
(2) FIG. **1** is a diagram showing an energy storage system (ESS) according to an embodiment of the present disclosure.
(3) FIG. **2** is a diagram for illustrating the connection structure between a water tank and a battery

module and the relationship between the water tank and a controller, in a battery pack according to an embodiment of the present disclosure.
(4) FIG. **3** is a diagram for illustrating the relationship among a sensor, the controller and the water tank, in the battery pack according to an embodiment of the present disclosure.
(5) FIGS. **4** and **5** are perspective views showing a battery module applied to the battery pack according to an embodiment of the present disclosure.
(6) FIG. **6** is a diagram showing an inner structure of the battery module applied to the battery pack according to an embodiment of the present disclosure.
(7) FIGS. **7** to **11** are diagrams showing a detailed configuration of a coolant tube applied to the present disclosure.
(8) FIG. **12** is a diagram showing an expansion pad applied to the battery pack according to an embodiment of the present disclosure.
(9) FIG. **13** is a diagram showing a battery pack according to another embodiment of the present disclosure to which a valve is applied.
(10) FIG. **14** is a diagram showing a battery pack according to still another embodiment of the present disclosure to which a barrier opened by heat is applied.
BEST MODE
(11) Hereinafter, preferred embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. Prior to the description, it should be understood that the terms used in the specification and the appended claims should not be construed as limited to general and dictionary meanings, but interpreted based on the meanings and concepts corresponding to technical aspects of the present disclosure on the basis of the principle that the inventor is allowed to define terms appropriately for the best explanation. Therefore, the description proposed herein is just a preferable example for the purpose of illustrations only, not intended to limit the scope of the disclosure, so it should be understood that other equivalents and modifications could be made thereto without departing from the scope of the disclosure.
(12) Referring to FIG. **1**, an energy storage system (ESS) according to an embodiment of the present disclosure includes a plurality of battery packs **100** according to an embodiment of the present disclosure.
(13) Referring to FIGS. **1** to **3**, the battery pack **100** according to an embodiment of the present disclosure includes a pack housing **110**, a battery module **120**, a water tank **130**, a controller **140**, a coolant tube **150** and a sensor **160**.
(14) The pack housing **110** is an approximately rectangular frame that defines the appearance of the battery pack **100**, and has a space formed therein so that the plurality of battery modules **120**, the water tank **130**, the controller **140**, the coolant tube **150** and the sensor **160** may be installed therein.
(15) The battery module **120** is provided in plural, and the plurality of battery modules **120** are vertically stacked in the pack housing **110** to form a single module stack. The specific structure of the battery module **120** will be described later in detail with reference to FIGS. **4** to **6**.
(16) The water tank **130** is provided inside the pack housing **110** and stores a coolant that is to be supplied to the battery module **120** when a thermal runaway phenomenon occurs at the battery module **120**. The water tank **130** may be disposed above the module stack for quick and smooth supply of coolant. In this case, even though a separate coolant pump is not used, the coolant may be rapidly supplied to the battery module **120** by free fall and water pressure of the coolant. Of course, a separate coolant pump may also be applied to the water tank **130** to supply the coolant more quickly and smoothly. In this case, if a coolant pump has a sufficient pressure, the water tank **130** may be disposed at the same height as the module stack or at a lower position.
(17) The controller **140** may be connected to the sensor **160** and the water tank **130** and output a control signal to open the water tank **130** and/or a control signal to operate the coolant pump according to the sensing signal of the sensor **160**. In addition, the controller **140** may additionally

perform a function as a battery management system (BMS) that is connected to each battery module **120** to manage charging and discharging thereof, in addition to the above function.

(18) The controller **140** outputs a control signal to open the water tank **130** when detecting gas or a temperature rise above a reference value inside the battery pack **100** due to a thermal runaway phenomenon occurring in at least one of the plurality of battery modules **120**, and allows the coolant to be supplied to the battery module **120** accordingly.

(19) If the water tank **130** is opened according to the control signal of the controller **140**, the coolant is sequentially supplied from a battery module **120** located at an upper portion to a battery module **120** located at a lower portion. Thus, the flame in the battery modules **120** is extinguished and also the battery modules **120** are cooled, thereby preventing the thermal runaway phenomenon from spreading throughout the battery pack **100**.

(20) The coolant tube **150** connects the water tank **130** and the battery module **120** to each other, and functions as a passage for carrying the coolant supplied from the water tank **130** to the battery module **120**. To perform this function, one end of the coolant tube **150** is connected to the water tank **130**, and the other end of the coolant tube **150** is branched by the number of the battery modules **120** and connected to the plurality of battery modules **120**, respectively. Detail structure of the coolant tube **150** will be described later with reference to FIGS. **7** to **11**.

(21) If the thermal runaway phenomenon occurs in at least a part of the plurality of battery modules **120** as described above, the sensor **160** detects a temperature rise and/or a gas ejection and transmits a detection signal to the controller **140**. To perform this function, the sensor **160** may be a temperature sensor or a gas detection sensor, or a combination of the temperature sensor and the gas detection sensor.

(22) The sensor **160** is installed inside the pack housing **110** to detect temperature rise or gas generation inside the battery pack **100**. The sensor **160** may be attached to an inner side or outer side of each of the plurality of battery modules **120** to quickly sense the temperature of the battery module **120** and/or gas generated from the battery module **120**.

(23) Next, the battery module **120** applied to the battery pack **100** according to an embodiment of the present disclosure will be described in more detail with reference to FIGS. **4** and **6** to **12**.

(24) Referring to FIGS. **4** and **6** to **12**, the battery module **120** may be implemented to include a plurality of battery cells **121**, a bus bar frame **122**, a module housing **123**, an air inlet **124** and an air outlet **125**. In addition, referring to FIG. **12**, the battery module **120** according to an embodiment of the present disclosure may further include an expansion pad **126** additionally.

(25) The battery cell **121** is provided in plural, and the plurality of battery cells **121** are stacked to form one cell stack. As the battery cell **121**, for example, a pouch-type battery cell may be applied. The battery cell **121** includes a pair of electrode leads **121***a* respectively drawn out at both sides thereof in a longitudinal direction.

(26) The bus bar frame **122** is provided in a pair, and the pair of bus bar frames **122** cover one side and the other side of the cell stack in a width direction. The electrode lead **121***a* of the battery cell **121** is drawn through a slit formed at the bus bar frame **122**, and is bent and fixed by welding or the like onto the bus bar frame **122**. That is, the plurality of battery cells **121** may be electrically connected by the bus bar frame **122**.

(27) The module housing **123** has a substantially rectangular parallelepiped shape, and accommodates the cell stack therein. The air inlet **124** and the air outlet **125** are formed at one side and the other side of the module housing **123** in the longitudinal direction.

(28) The air inlet **124** is formed at one side of the cell stack in the stacking direction, namely at one side of the battery module **120** in the longitudinal direction and has a hole shape formed through the module housing **123**. The air outlet **125** is formed at the other side of the cell stack in the stacking direction, namely at the other side of the battery module **120** in the longitudinal direction and is has a hole shape formed through the module housing **123**.

(29) The air inlet **124** and the air outlet **125** are located at diagonally opposite sides along the

longitudinal direction of the battery module **120**.
(30) Meanwhile, an empty space is formed between the bus bar frame **122** and the module housing **123**. That is, the empty space in which air for cooling the battery cell **121** flows is formed between one of six outer surfaces of the module housing **123** facing one side and the other side of the battery cell **121** in the longitudinal direction and the bus bar frame **122**. The empty space is formed at each of both sides of the battery module **120** in the width direction.
(31) The air inlet **124** is formed at a location corresponding to the empty space formed at one side of the battery module **120** in the width direction, and the air outlet **125** is formed at a location corresponding to the empty space formed at the other side of the battery module **120** in the width direction.
(32) In the battery module **120**, the air introduced therein through the air inlet **124** cools the battery cell **121** while moving from the empty space formed at one side of the battery module **120** in the width direction to the empty space formed at the other side of the battery module **120** in the width direction, and then goes out through the air outlet **125**. That is, the battery module **120** corresponds to an air-cooled battery module.
(33) Referring to FIG. **12**, the expansion pad **126** is disposed inside the air inlet **124** and the air outlet **125**, and has a size smaller than the opened area of the air inlet **124** and the air outlet **125**. The expansion pad **126** preferably has a size of less than about 30% compared to the opened area of the air inlet **124** and air outlet **125** so that the air may flow through the air inlet **124** and the air outlet **125** smoothly when the battery module **120** is normally used.
(34) The expansion pad **126** is expanded by contacting the coolant introduced into the battery module **120** to close the air inlet **124** and the air outlet **125**. The expansion pad **126** contains a resin that exhibits a very large expansion rate when absorbing moisture, for example a resin that increases in volume by at least about two times or more compared to the initial volume when a sufficient amount of moisture is provided thereto. As a resin used for the expansion pad **126**, a polyester staple fiber is mentioned, for example.
(35) By applying the expansion pad **126**, when a thermal runaway phenomenon occurs in at least some battery modules **120** and thus a coolant is introduced into the battery modules **120**, the air inlet **124** and the air outlet **125** are closed. If the air inlet **124** and the air outlet **125** are closed as above, the coolant introduced into the battery module **120** does not escape to the outside but stays inside the battery modules **120**, thereby quickly resolving the thermal runaway phenomenon occurring in the battery modules **120**.
(36) Next, the coolant tube **150** applied to the battery pack **100** according to an embodiment of the present disclosure will be described in more detail with reference to FIGS. **7** to **11** along with FIGS. **2** and **6**.
(37) Referring to FIGS. **7** to **11** along with FIGS. **2** and **6**, the coolant tube **150** includes a main tube **151**, a plurality of supply tubes **152**, and a bypass tube **153**. In addition, the coolant tube **150** may further include at least one sub bypass tube **154** additionally.
(38) The main tube **151** is directly connected to the water tank **130**. The main tube **151** has a shape extending downward from the water tank **130** along a stacking direction of the module stack.
(39) The plurality of supply tubes **152** are branched from the main tube **151** to connect the battery modules **120** to each other. The supply tube **152** is provided in the same number as the number of battery modules **120**. The supply tube **152** passes through the module housing **123** from one side or the other side of the cell stack in the stacking direction and communicates with an empty space formed between the bus bar frame **122** and the module housing **123**. That is, the supply tube **152** may be inserted through the same surface as the surface in which the air inlet **124** is formed or the surface in which the air outlet **125** is formed, among six surfaces of the module housing **123**.
(40) Therefore, as shown in FIGS. **4** and **5**, the coolant introduced into the battery module **120** through the supply tube **152** flows from the empty space formed at one side of the battery module **120** in the width direction to the empty space formed at the other side of the battery module **120** in

the width direction and fills the inside of the battery module **120**.
(41) The bypass tube **153** is connected to the main tube **151**. The bypass tube **153** connects one longitudinal side and the other longitudinal side of the main tube **151**. That is, the bypass tube **153** and the main tube **151** are connected at two points. A first connection point is located above a supply tube **152** located at a top end along the stacking direction of the module stack, and a second connection point is located below a supply tube **152** located at a bottom end along the stacking direction of the module stack. Due to the bypass tube **153**, the battery pack **100** according to the present disclosure may supply the coolant from at the upper and lower portions of the module stack at the same time, thereby minimizing the occurrence of deviation in the amount of coolant supplied to the plurality of battery modules **120**.
(42) Meanwhile, the sub bypass tube **154** connects the bypass tube **153** and the main tube **151** between both ends of the bypass tube **153**. When the plurality of supply tubes **152** are grouped into a plurality of supply tube groups along a direction from top to bottom and each group includes at least two supply tubes **152**, the sub bypass tube **154** may be respectively connected between adjacent supply tube groups.
(43) Referring to FIG. **8**, a case where two supply tube groups are formed by grouping six supply tubes **152** into one supply tube group along a direction from top to bottom and the sub bypass tube **154** is connected between the two supply tube groups is illustrated.
(44) Referring to FIG. **9**, a case where three supply tube groups are formed by grouping four supply tubes **152** into one supply tube group along a direction from top to bottom and the sub bypass tube **154** is respectively connected between adjacent supply tube groups is illustrated.
(45) Referring to FIG. **10**, a case where four supply tube groups are formed by grouping three supply tubes **152** into one supply tube group along a direction from top to bottom and the sub bypass tube **154** is respectively connected between adjacent supply tube groups is illustrated.
(46) Referring to FIG. **11**, a case where six supply tube groups are formed by grouping two supply tubes **152** into one supply tube group along a direction from top to bottom and the sub bypass tube **154** is respectively connected between adjacent supply tube groups is illustrated.
(47) The examples shown in FIGS. **8** to **11** show only the cases where each supply tube group is formed with the same number of supply tubes **152**, but the present disclosure is not limited thereto, and a case where different supply tube groups have a different number of supply tubes **152** also falls within the scope of the present disclosure.
(48) Next, a battery pack according to another embodiment of the present disclosure will be described with reference to FIG. **13**.
(49) The battery pack according to another embodiment of the present disclosure is different from the battery pack **100** according to an embodiment of the present disclosure described above only in the point that a valve **170** is installed inside the supply tube **152**, and other components are substantially the same.
(50) Thus, the battery pack according to another embodiment of the present disclosure will be described based on the valve **170**, and the features identical to the former embodiment will not be described in detail.
(51) The valve **170** is provided in plural as much as the number of the plurality of battery modules **120**, and the valves **170** are respectively installed adjacent to the plurality of battery modules **120** to individually allow or block the flow of coolant flowing into the plurality of battery modules **120**.
(52) As described above, in order to operate the plurality of valves **170** independently, at least one sensor **160** may be provided for each battery module **120**. Thus, if the sensor **160** is provided for every battery module **120**, it is possible to input the coolant only to some battery modules **120** in which a thermal runaway phenomenon occurs.
(53) That is, if the controller **140** receives a detection signal from some sensors **160**, the controller **140** determines that a thermal runaway phenomenon occurs in the battery modules **120** to which the sensors **160** sending the detection signal are attached, and opens the valves **170** installed

adjacent to the battery modules **120** where the thermal runaway phenomenon occurs among the plurality of valves **170** so that the coolant may be put thereto.

(54) Next, a battery pack according to still another embodiment of the present disclosure will be described with reference to FIG. **14** along with FIG. **6**.

(55) The battery pack according to still another embodiment of the present disclosure differs from the battery pack **100** according to another embodiment of the present disclosure, described above, in that a barrier **180** is installed in the supply tube **152** instead of the valve **170**, and other components are substantially the same.

(56) Therefore, in describing the battery pack according to still another embodiment of the present disclosure, the barrier **180** will be described in detail, and other components substantially the same as the former embodiment will not be described in detail.

(57) The barrier **180** is installed inside each of the plurality of supply tubes **152** and blocks the coolant from being supplied to the battery module **120** during normal use of the battery pack **100** and the ESS. The barrier **180** is broken when the temperature of the battery module **120** rises and exceeds a reference temperature, so that the coolant may be supplied into the battery module **120** through the supply tube **152**.

(58) The barrier **180** may be made of a resin film, and its thickness and specific material may be determined in consideration of the capacity and number of battery cells **121** included in the battery module **120**.

(59) As described above, the battery pack according to the present disclosure may prevent a thermal runaway phenomenon from spreading to neighboring battery modules **120** by putting a coolant into the battery module **120** when the thermal runaway phenomenon occurs in the battery module **120**. In particular, by applying the bypass tube **153**, when the thermal runaway phenomenon occurs in the plurality of battery modules **120**, the battery pack according to the present disclosure may uniformly supply the coolant to every battery module **120**, which allows the cooling of the battery module **120** relatively far away from the water tank **130** not to be weakened. In addition, since the battery pack according to the present disclosure has a structure in which the air inlet **124** and the air outlet **125** may be closed when a coolant is inserted into the air-cooled battery module **120** so that the coolant may be filled therein, it is possible to more effectively prevent the thermal runaway phenomenon from spreading.

(60) The present disclosure has been described in detail. However, it should be understood that the detailed description and specific examples, while indicating preferred embodiments of the disclosure, are given by way of illustration only, since various changes and modifications within the scope of the disclosure will become apparent to those skilled in the art from this detailed description.

Claims

1. A battery pack, comprising: a pack housing; a plurality of battery modules stacked in the pack housing; a water tank connected to the plurality of battery modules and configured to store a coolant; a coolant tube including a main tube connected to the water tank, a plurality of supply tubes configured to connect the main tube to each of the battery modules, and a bypass tube connected to the main tube at a first point above a supply tube located at a top end among the plurality of supply tubes and at a second point below a supply tube located at a bottom end among the plurality of supply tubes, wherein the main tube and the bypass tube form a loop, and wherein the plurality of supply tubes connect to a portion of the main tube forming the loop; at least one sensor installed in the pack housing to detect a thermal runaway phenomenon generated in at least a part of the plurality of battery modules; and a controller configured to output a control signal for introducing a coolant into the battery module through the coolant tube when a thermal runaway phenomenon is detected by the at least one sensor.

2. The battery pack according to claim 1, wherein each of the plurality of battery modules includes: a cell stack formed by stacking a plurality of battery cells; a module housing configured to accommodate the cell stack; an air inlet formed through the module housing at a first side of the cell stack in a stacking direction; and an air outlet formed through the module housing at a second side of the cell stack in the stacking direction.
3. The battery pack according to claim 2, wherein each of the plurality of battery modules includes an expansion pad disposed inside each of the air inlet and the air outlet and configured to expand due to contact with the coolant introduced into the battery module to close the air inlet and the air outlet.
4. The battery pack according to claim 2, wherein each of the plurality of battery modules includes a pair of bus bar frames respectively coupled to the first side and the second side of the cell stack.
5. The battery pack according to claim 4, wherein the air inlet and the air outlet are formed at locations corresponding to an empty space formed between the bus bar frame and the module housing.
6. The battery pack according to claim 4, wherein the supply tube passes through the module housing from the first side of the cell stack to the second side and communicates with an empty space formed between the bus bar frame and the module housing.
7. The battery pack according to claim 2, wherein the battery pack includes a plurality of valves respectively installed in the plurality of supply tubes, and wherein the plurality of valves are respectively installed adjacent to the plurality of battery modules to individually allow or block the flow of coolant flowing into the plurality of battery modules.
8. The battery pack according to claim 7, wherein the at least one sensor is installed to each of the plurality of battery modules.
9. The battery pack according to claim 8, wherein the controller outputs the control signal to open a valve installed adjacent to a battery module of the plurality of battery modules where a thermal runaway phenomenon is detected by the at least one sensor.
10. The battery pack according to claim 2, further comprising a barrier installed in each of the plurality of supply tubes.
11. The battery pack according to claim 10, wherein the barrier is broken when a temperature of the battery module exceeds a reference temperature.
12. An energy storage system (ESS), comprising a plurality of battery packs according to claim 1.
13. The battery pack according to claim 1, further comprising at least one sub bypass tube extending between the main tube and the bypass tube at a point between the first point and the second point.
14. The battery pack according to claim 1, further comprising a plurality of sub bypass tubes extending between the main tube and the bypass tube between the first point and the second point.
15. The battery pack according to claim 14, wherein an equal number of supply tubes of the plurality of supply tubes is located between each pair of adjacent sub bypass tubes of the plurality of sub bypass tubes.